RV College of Engineeri
Myscre Road, RV Vidyaniketan Post,
Bengaluru - 560059, Kamataka, India

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

Go, change the world

V SEMESTER

Experiential Learning - ODD 2025-26

Phase-I Review

on

Theme: AIML

Prediction of Infrastructure Failures using ML Model

Team Members

Raju Dhangar	Hamaas Shabir	Harsh Dubey	Rahul Laxman
1RV23CS194	1RV23IS052	1RV23IS053	1RV23CS189
dhangarr.cs23@rvce.edu.in	hamaass.is23@rvce.edu.in	harshdubey.is23@rvce.edu.in	rahullaxmanr.cs23@rvce.edu.in

MENTOR

Dr. Praveen Kumar

Professor

Department of Civil Engineering

RV College of Engineering

Bengaluru-59

Contents

1. Introduction
2. Existing system
3. Identification of problem
4. Literature Review
5. Problem definition
6. Objectives
7. Methodology
8. Proposed tools and techniques.
9. Expected outcome.
10. Conclusion
11. References

Introduction

- Modern infrastructure such as buildings, bridges, and industrial assets undergo **continuous stress** from **environmental**, **mechanical**, and **operational loads**.
- Unexpected failures lead to **safety hazards**, **economic loss**, and **downtime**.
- Structural Health Monitoring (SHM) uses sensors (accelerometers, strain gauges, temperature sensors) to **capture real-time physical behavior.**
- Any deviation in sensor value is automatically classified as: **Healthy, Minor Deviation, or Potential Damage**, enabling early alerts before critical failure.

- This project delivers a **sensor-aware, edge-deployable ML system** capable of continuous, real-time infrastructure health prediction and anomaly detection.

Existing Systems

Traditional infrastructure monitoring relies on:

- **Manual inspections** (slow, subjective, labor-intensive,on site engineers, manual threshold setting)
- **Periodic structural audits** (not real-time)
- **Conventional threshold-based alert systems** (Principal Component Analysis,Statistical Process Control (SPC))
- **Primarily Signal Processing** and **Statistical Model Updating**
- **No predictive analytics** → Failures detected only after they occur

Limitations:

- Not scalable
- Not cost effective
- No continuous monitoring

Identification of problem

Infrastructure failures often show **micro-level symptoms** **long before visible cracks** appear. Current systems **capture** subtle **vibration/strain patterns** but use signal processing only!

There is **no low-cost**, **intelligent** SHM(Structural Health Monitoring)

OUR SYSTEM combining:

- Multi-sensor data (accelerometers, strain gauges, temperature)
- Time-series feature learning using a **pretrained 1D CNN encoder**
- Real-time fine tuning of **LSTM decoder** and **anomaly scoring** through an edge-deployable autoencoder model

Thus, there is a **need** for a **predictive**, **automated**, **ML-based system** capable of early damage detection and continuous **structural resilience monitoring.**

Objectives

- To design a **low-cost** **Structural Health Monitoring (SHM) system** using accelerometer, strain gauge, and temperature sensors.
- To **collect** and **preprocess** **time-series sensor data** for **accurate** analysis of structural behavior.
- To extract meaningful features from **vibration**, **strain**, and **environmental patterns** for **damage assessment**.
- To develop and train machine learning models to classify structural conditions as **Healthy**, **Minor** Damage, or **Severe** Damage.
- To enable **early detection** of structural deterioration and support predictive maintenance aligned with **SDG 9**.

Hardware Bill of Materials (BOM)

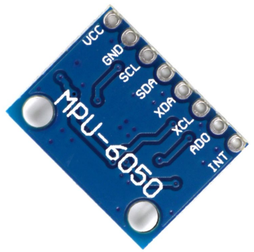
Component Category	Part / Specification	Cost (₹)
Microcontroller	ESP32	400
Accelerometer	ADXL345	200
Temp. Sensor	DS18B20	75
Strain Gauge	BF350	30
Amplifier	HX711	40
Battery(8-9 hours)	LiPo 2500mAh	600
Power Mgmt	TP4056	20
Voltage Reg	Boost Converter	40
Lighting	5V LED Strips	500
Housing	Hard Case Enclosure	500
TOTAL		**₹2405**

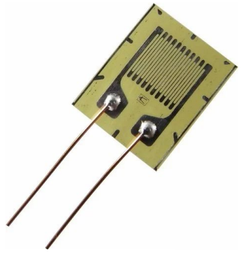
Proposed tools and techniques

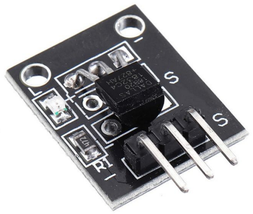
Go, change the world

Hardware:

- ESP32 microcontroller
- ADXL355 accelerometer
- Foil strain gauge
- DS18B20 temperature sensor







Software & ML Tools:

- Python, NumPy, Pandas
- TensorFlow / Keras for ML
- Google Collab
- MATLAB (SHM signal analysis)
- Streamlit/Dashboard for visualization

Other Tools:

- Soldering equipment, data logger, adhesives

Expected Outcomes

Go, change the world

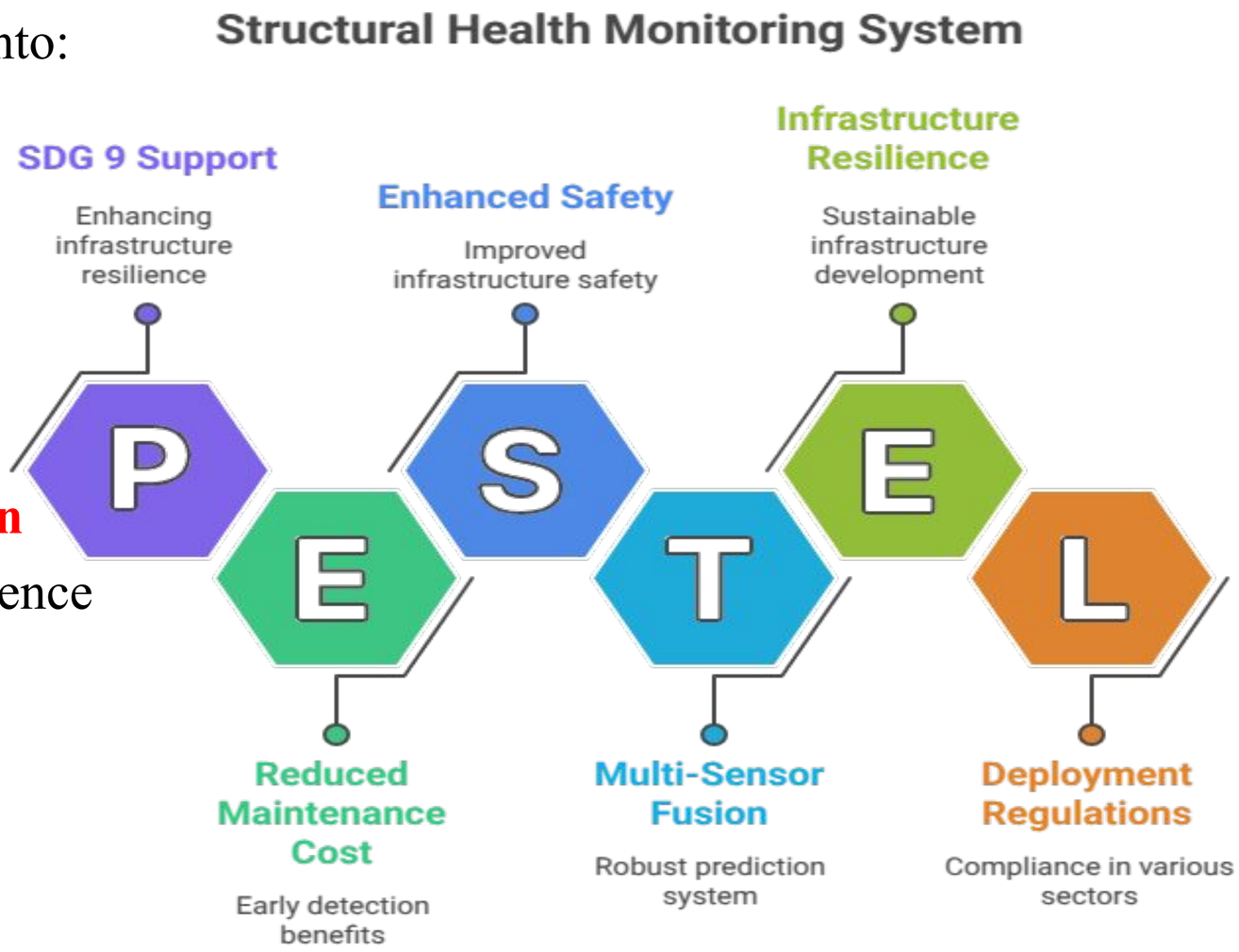
- **Accurate classification** of structural conditions into:
  - Healthy (0)
  - Minor Damage (1)
  - Severe Damage (2)
- Multi-sensor fusion → more robust prediction
- **Low-cost**, scalable prototype for SHM
- **Reduced maintenance cost** due to **early detection**
- Supports **SDG 9** by enhancing infrastructure resilience
- Potential deployment in:

  >Buildings

  >Footbridges

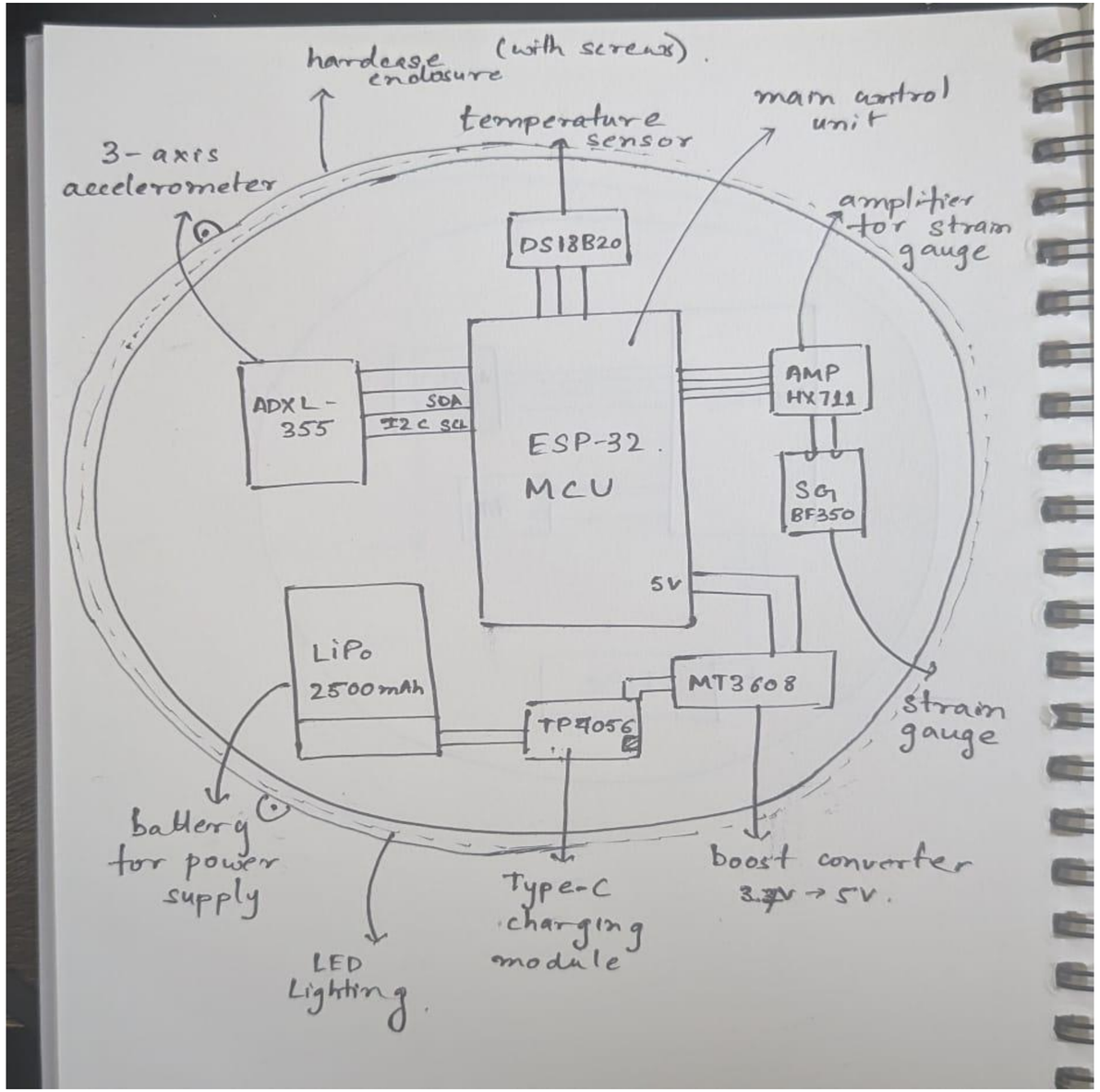
  >Industrial plants

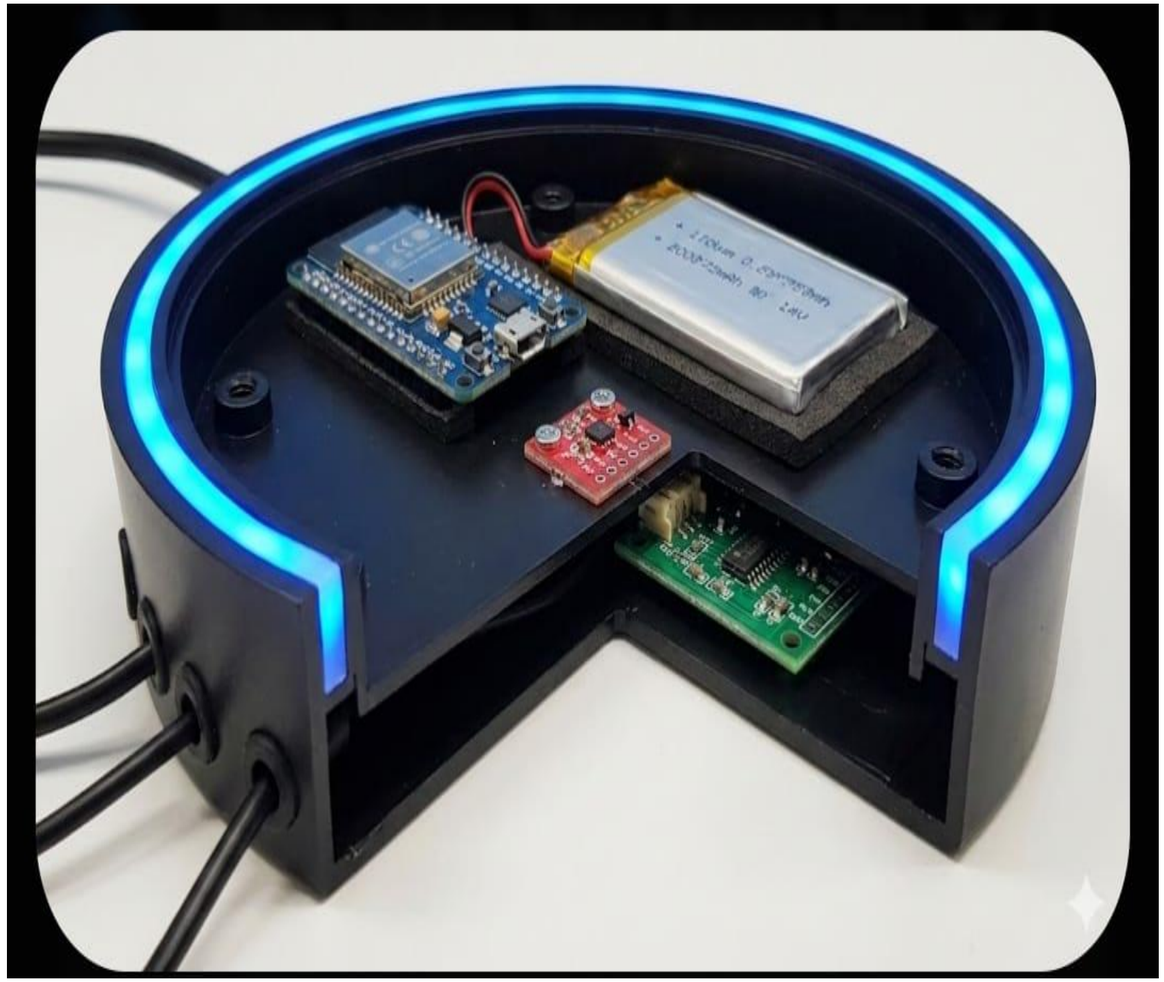
  >Towers



Expected Outcomes

Go, change the world





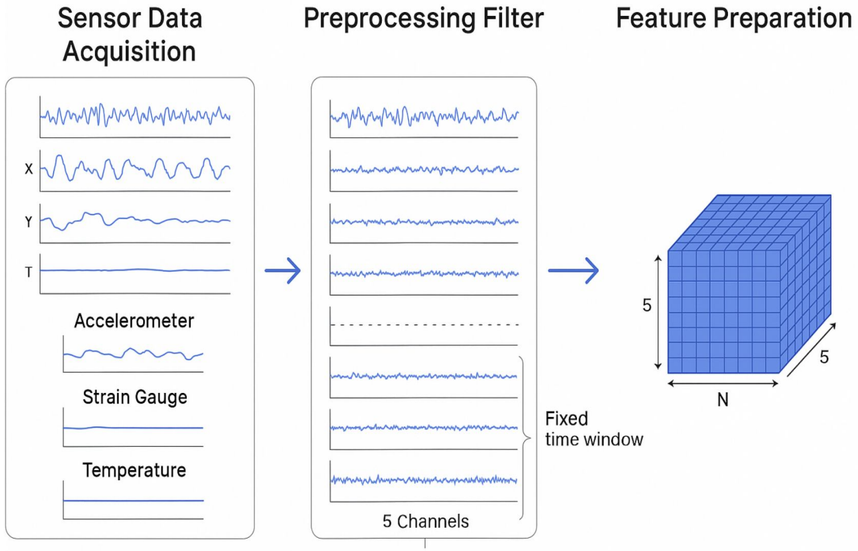
Methodology

Go, change the world

- **1.Sensor Data Acquisition:** Collect high-precision, synchronized **time-series data** from the Accelerometer (X, Y, Z axes), Strain Gauge, and Temperature sensors.

- **2.Preprocessing Filter:** Filter noise (especially from strain/ADC), normalize all signals, and **segment data** into fixed time windows (1–3 seconds), ensuring all 5 channels are perfectly synchronized withi the window.

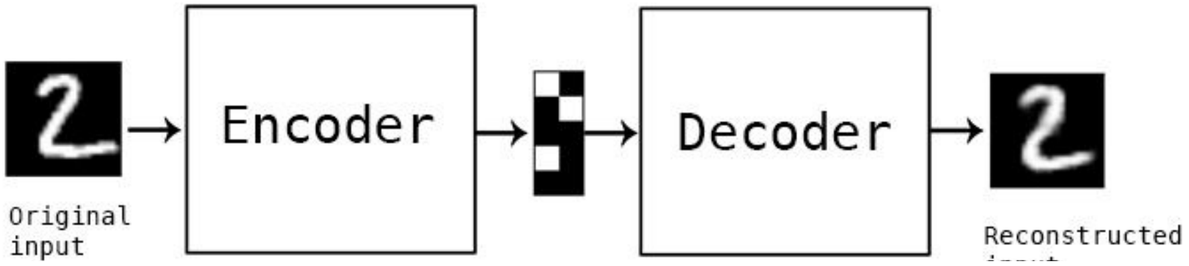
- **3.Feature Preparation:** arranging your time-series sensor windows so **CNN or LSTM** can learn patterns from them.Prepare **3D Input Tensor** for the 1D CNN. The input shape for a time window will be: **(Samples, Timestamps, Features) -> (1, N, 5).** This feeds the full sequence to the 1D CNN for local feature extraction.

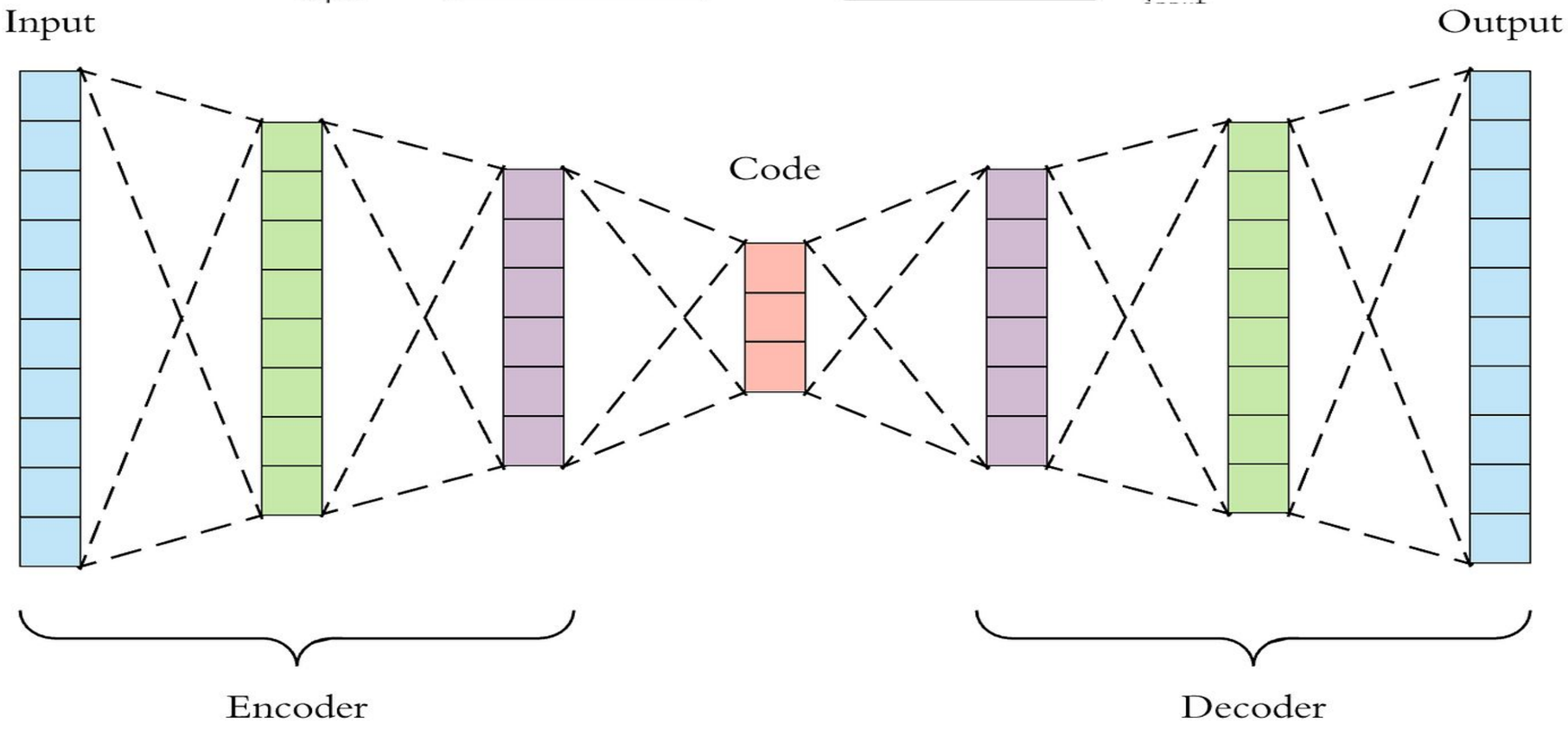


Methodology

Go, change the world

Reconstruction Error $E_t^{(n)} = ||\mathbf{x}_t^{(n)} - \hat{\mathbf{x}}_t^{(n)}||_2^2$





Methodology

ML Modeling Pipeline

Pretraining (Offline)

- **Phase 1: Universal Encoder Pre-training** (Cloud): Train a 1D CNN Encoder (connected to a temporary 1D CNN Decoder) on large, diverse, healthy datasets.
- LANL SHM dataset | Z24 bridge dataset | IASC-ASCE dataset|UBC lab beams
- The CNN learns **general local feature patterns** (spikes, frequencies).Learns to generalize SHM vibration patterns.

Fine-tuning on New Building

- Freeze encoder
- **Phase 2: Fine-Tuning** (Cloud): Freeze the 1D CNN Encoder. Attach a new LSTM Decoder head and train only its weights using the healthy data from the target structure. The LSTM learns the building-specific temporal sequence reconstruction logic.

Edge deployment

- **Optimize and Quantize** the full Hybrid CNN-LSTM Autoencoder model (e.g., FLOAT32 to INT8 using **TFLite-Micro**). Deploy the optimized model to the ESP32 edge device for low-latency, real-time inference

Methodology

Go, change the world

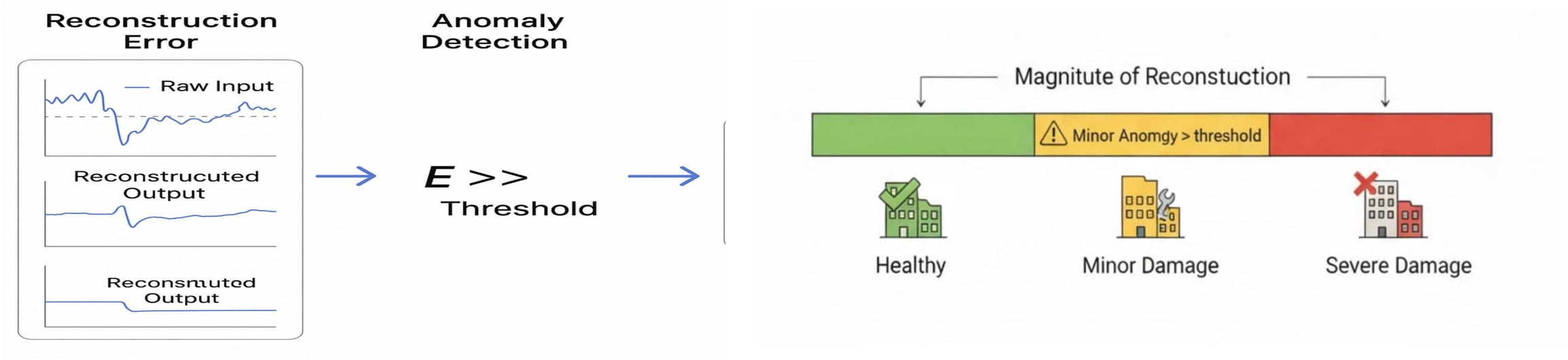
$$A_i = \|x_i - \hat{x}_i\|^2$$

Anomaly Detection & Classification:

Phase 3 Reconstruction Error: On the ESP32, calculate the error (E) between the raw_input and the model's reconstructed output.

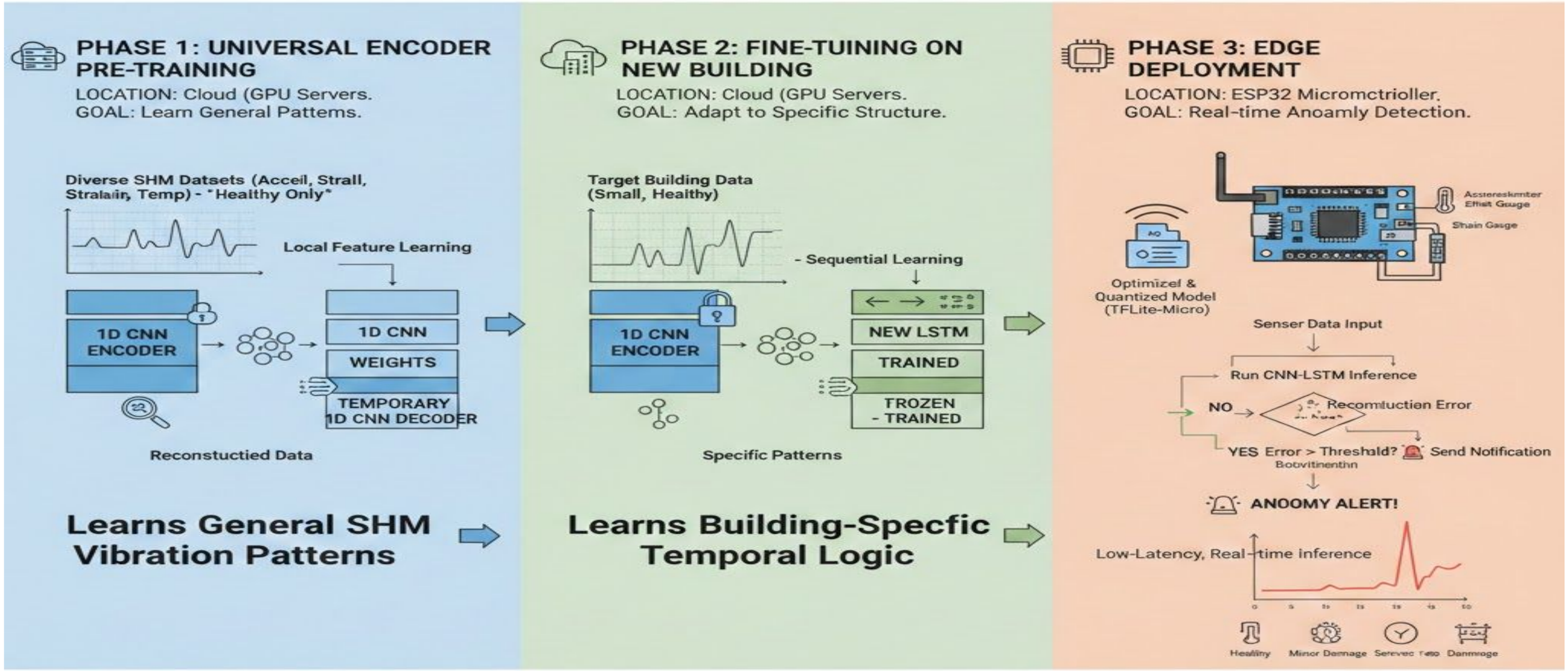
Anomaly Detection: Flag an alert if **E >>Threshold.**

Damage Classification: Use the magnitude of E or a simple post-processing classifier to categorize the anomaly: **Healthy, Minor Anomaly, or Severe Anomaly.**



Methodology

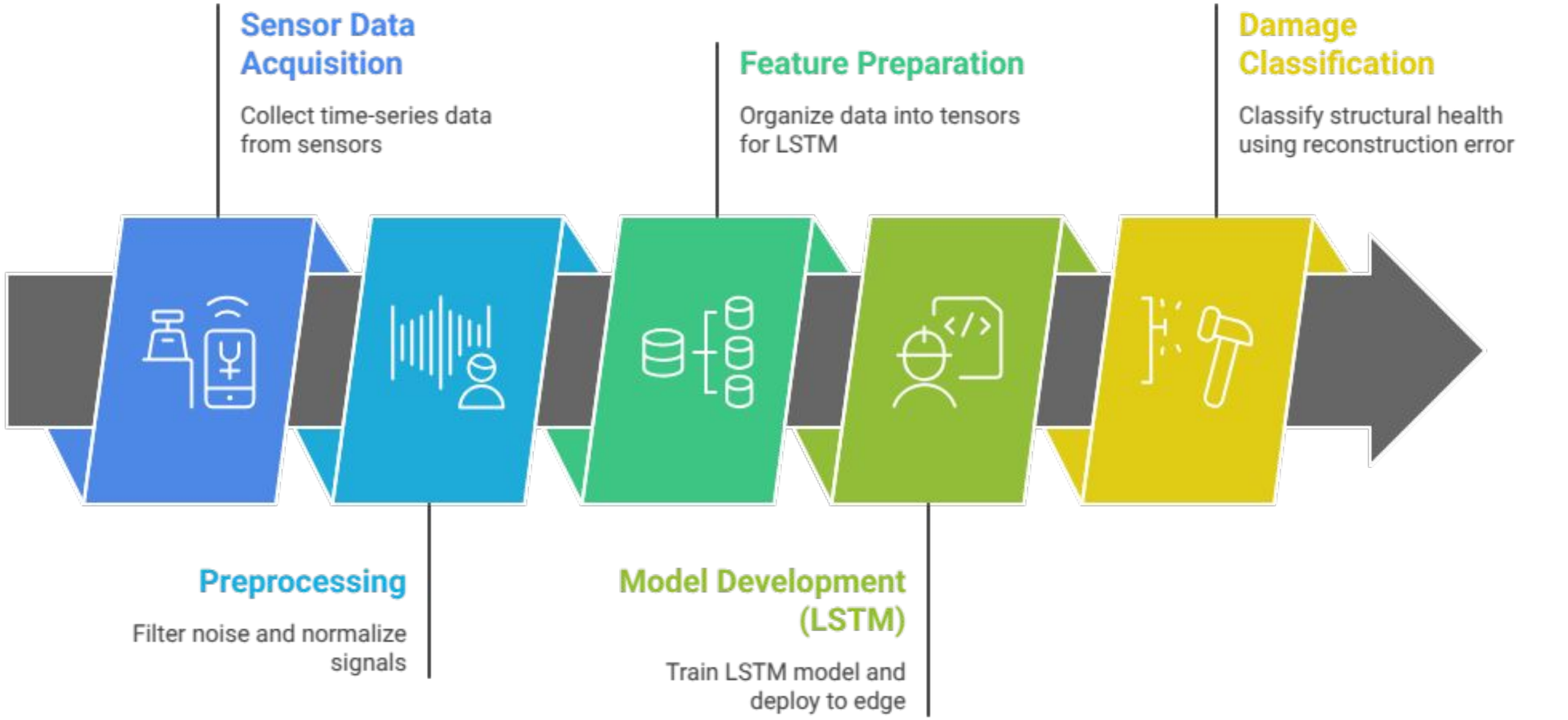
Go, change the world



Methodology

Go, change the world

Structural Health Monitoring Process



Made with Napkin

Expected Outcomes

Go, change the world

#1 Novelty : A Universal SHM Encoder pretrained

1.Datasets from multiple online SHM datasets Sources:

LANL SHM dataset | Z24 bridge dataset | IASC-ASCE dataset|UBC lab beams

2.How the Universal Encoder is Trained

Build an autoencoder (Encoder + temporary decoder)

Train on all pretraining data → learns to compress & reconstruct

✔ **Encoder = "global knowledge"**

✔ **Decoder = "building-specific knowledge"**

Hybrid CNN-LSTM Models

Function: This combination is highly effective. A 1D-CNN layer is placed first (the encoder part) to rapidly extract essential local features (like spikes, short patterns, or transients) from the time-series window.

->Save encoder.h5->Discard pretrained decoder

The encoder now contains core SHM feature understanding:

- Common vibration/strain shapes & frequency bands
- Cross-axis correlations (X–Y–Z)
- Typical healthy structural behavior patterns
- Robustness to noise, drift, and sensor variability

3.Freeze universal encoder->Train a new lightweight LSTM decoder, which process the sequence of extracted features over time decoder using only healthy data from the target building->**Creates a self-calibrating anomaly detection system**

Expected Outcomes

Go, change the world

#2 Novelty : Distributed SHM Nodes for Voting & Damage Localization

1.Multiple Low-Cost SHM Nodes

Each individual structure zone has several sensor node (accelerometer + strain + temperature)

Each node runs **local CNN-LSTM-AE anomaly scoring** on the edge

$$\text{Reconstruction Error } E_t^{(n)} = ||\mathbf{x}_t^{(n)} - \hat{\mathbf{x}}_t^{(n)}||_2^2$$

2.Collaborative Anomaly Voting

Nodes share anomaly scores with neighbors (ESP-NOW)

A structural region is flagged only if multiple nodes agree

Reduces false positives and improves reliability(binary voting)

$$A_n = \begin{cases} 1 & \text{if } S_n \geq T_{local} \\ 0 & \text{otherwise} \end{cases}$$

$$R = \begin{cases} 1 & \text{if } C \geq T_{vote} \\ 0 & \text{otherwise} \end{cases}$$

$$C = \sum_{m \in \mathcal{N}_{active}} A_m$$

$$T_{\text{local}}^{(n)} = \mu_{E_h}^{(n)} + \beta \cdot \sigma_{E_h}^{(n)}$$

$$P(A_n = 1 \text{ and } A_m = 1) = P_{FP} \times P_{FP} \approx 0.003 \times 0.003 = 0.000009$$

Expected Outcomes

#2 Novelty : Distributed SHM Nodes for Voting & Damage Localization

3.Triangulation-Based Damage Localization via MESH NETWORK

When several nearby nodes detect elevated anomaly scores(avg voting):

- Combine their **spatial positions(x,y,z)**
- Use score intensity (Sn)+ timing correlation
- Estimate **probable damage zone**
- Enables localization without high-end equipment

$$x_{\text{damage}} = \frac{\sum_n S_n \cdot x_n}{\sum_n S_n} \qquad y_{\text{damage}} = \frac{\sum_n S_n \cdot y_n}{\sum_n S_n} \qquad z_{\text{damage}} = \frac{\sum_n S_n \cdot z_n}{\sum_n S_n}$$

Expected Outcomes

Expected Outcomes

Conclusion

Go, change the world

- The **pretrained 1D CNN Encoder + edge-LSTM decoder** enables far more precise and early detection of structural anomalies compared to traditional threshold-based SHM methods.
- Multi-sensor **data fusion** (**accelerometer (X,Y,Z)**, **strain**, **temperature**) improves **reliability**.
- This solution contributes significantly to sustainable, safe, and smart infrastructure systems in alignment with **SDG 9(Industry, Innovation, and Infrastructure)** by enabling proactive maintenance and resilience.
- **The prototype is easily extendable to:**
- **Real-time cloud dashboards** and remote monitoring
- IoT-enabled large-scale deployments due to cheap cost across multiple buildings/bridges
- Continuous **on-device fine-tuning** using **edge computing** (ESP32)
- **Adaptive anomaly scoring** for mesh

This is the AI-powered, Edge-optimized foundation for mass-scale, sustainable Structural Health Monitoring across the globe!

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Literature Review

Go, change the world

Sno	Paper Title	Authors	Main Contribution	Year
1.	Adversarially-Trained Tiny Autoencoders for Near-Sensor Continuous Structural Health Monitoring	Alessio Burrello, Giacomo Sintoni & Davide Brunelli	This paper introduces extremely lightweight "**tiny autoencoders"** trained with adversarial techniques to enhance sensitivity to subtle structural changes while running directly on near-sensor edge hardware. By using **adversarial perturbations** during training, the model becomes more robust to noise and more responsive to micro-damage signals, enabling continuous real-time monitoring on low-power devices such as microcontrollers. The work demonstrates that carefully optimized, adversarially enhanced small models can outperform larger conventional autoencoders in early anomaly detection, making it highly suitable for scalable, **low-cost SHM deployments.**	2022
2.	Transferring self-supervised pre-trained models for SHM data anomaly detection with scarce labeled data	Mingyuan Zhou, Xudong Jian, Ye Xia, Zhilu Lai	Proposes using **self-supervised pretraining** on large, heterogeneous SHM time-series so encoders learn general vibration/strain representations that can be transferred and fine-tuned to new structures with little labeled data. Demonstrates improved generalization and faster adaptation to target sites. arXiv	2024
3.	Deep learning-based detection of structural damage using time-series data	Hung V. Dang, Mohsin Raza, Tung V. Nguyen, T. Bui-Tien & Huan X. Nguyen	This paper demonstrates the effectiveness of deep learning models—particularly C**NNs and LSTMs—for detecting structural damage** directly from raw vibration time-series data without handcrafted feature engineering. The proposed models achieve higher accuracy and earlier damage recognition compared to traditional SHM methods. The study validates that **deep time-series architectures** can successfully classify structural conditions across multiple damage levels, establishing deep learning as a powerful approach for practical SHM applications.	2021

References

1.A. Burrello, G. Sintoni, D. Brunelli and L. Benini, "Adversarially-Trained Tiny Autoencoders for Near-Sensor Continuous Structural Health Monitoring," 2022 IEEE 4th International Conference on Artificial Intelligence Circuits and Systems (AICAS), Incheon, Korea, Republic of, 2022, pp. 439-442, doi: 10.1109/AICAS54282.2022.9869952. keywords: {Training;Buildings;Bridge circuits;Transportation;Telecommunication traffic;Benchmark testing;Prediction algorithms},

2.Dang, Hung & Raza, Mohsin & Nguyen, Tung & Bui-Tien, Thanh & Nguyen, Huan. (2020). Deep learning-based detection of structural damage using time-series data. Structure and Infrastructure Engineering. 17. 1-20. 10.1080/15732479.2020.1815225.

3.https://doi.org/10.48550/arXiv.2412.03880

4.Yuan, F.-G.; Zargar, S.A.; Chen, Q.; Wang, S. Machine learning for structural health monitoring: Challenges and opportunities. Sens. Smart Struct. Technol. Civ. Mech. Aerosp. Syst. 2020, 11379, 1137903. [CrossRef] 2. Ibrahim, A.; Eltawil, A.; Na, Y.; El-Tawil, S. A Machine Learning Approach for Structural Health Monitoring Using Noisy Data Sets. IEEE Trans. Autom. Sci. Eng. 2020, 17, 900–908. [CrossRef]

5.Sivasuriyan, A.; Vijayan, D.; Górski, W.; Wodzy ´nski, Ł.; Vaverková, M.; Koda, E. Practical Implementation of Structural Health Monitoring in Multi-Story Buildings. Buildings 2021, 11, 263. [CrossRef] 4. Kong, X.; Cai, C.-S.; Hu, J. The State-of-the-Art on Framework of Vibration-Based Structural Damage Identification for Decision Making. Appl. Sci. 2017, 7, 497. [CrossRef]

6. Valinejadshoubi, M.; Bagchi, A.; Moselhi, O. Structural health monitoring of buildings and infrastructure. Struct. Health Monit. 2016, 1, 50371. 6. Farrar, C.R.; Worden, K. Structural Health Monitoring: A Machine Learning Perspective; Wiley: Hoboken, NJ, USA, 2012. 7. Rytter, A. Vibrational Based Inspection of Civil Engineering Structures. Ph.D. Thesis, Aalborg University, Aalborg, Denmark, 1993. 8. Escamilla-Ambrosio, P.; Liu, X.; Lieven, N.; Ramírez-Cortés, J.J.r. ANFIS-Wavelet Packet Transform Approach to Structural Health Monitoring. Ratio 2010, 10, 1.

Thank You